LSKF-YOLO: Large Selective Kernel Feature Fusion Network for Power Tower Detection in High-Resolution Satellite Remote Sensing Images

Chaojun Shi, *Member, IEEE*, Xian Zheng, *Student Member, IEEE*, Zhenbing Zhao, *Member, IEEE*, Ke Zhang, *Member, IEEE*, Zibo Su, and Qiaochu Lu

***Abstract*— With the rapid development of high-resolution satellite remote sensing observation technology, power tower detection based on satellite remote sensing images has become a key research focus for power intelligent inspection. However, the performance of power tower detection in satellite remote sensing images needs improvement due to complex backgrounds, and small and nonuniform target sizes. To address this, this article first constructs a multiscene high-resolution satellite remote sensing power tower dataset and then proposes the large selective kernel feature fusion (LSKF)-you only look once (YOLO) network for high-resolution satellite remote sensing images. This network primarily consists of a large spatial kernel selective attention fusion module and a multiscale feature alignment fusion (MFAF) structure. The large spatial selective kernel mechanism (LSKM) is improved by using the attentional feature fusion module, which provides richer feature information for accurately locating the position of the power tower. The MFAF structure effectively utilizes low-level semantic information, mitigates feature ambiguity in deeper network layers, and enables multiscale feature fusion of power towers within complex backgrounds. In addition, the introduction of minimum point distance-IoU (MPDIoU) enhances complete-IoU (CIoU), further improving the model's performance. The results demonstrate that the $F1$ score and mAP0.5 of the LSKF-YOLO network reach 0.764% and 77.47%, respectively. Compared with other deep learning-based satellite remote sensing power tower inspection methods, the LSKF-YOLO network significantly enhances detection accuracy and provides crucial technical support for intelligent inspection of power lines via satellite remote sensing.**

***Index Terms*— Feature alignment fusion, high-resolution satellite remote sensing images, large selective kernel mechanism, minimum point distance-IoU (MPDIoU), power tower detection.**

I. Introduction

With the ongoing development and application of artificial intelligence technology, power line inspection has shifted from digital to intelligent. Unmanned aerial vechicle (UAV) has become the primary method for power line inspection due to high efficiency, safety, and flexibility [1], [2]. However, UAV inspection is susceptible to weather changes and endurance issues, leading to insufficient macro power line monitoring capability and posing risks to the power grid's stable operation [3]. In recent years, satellite remote sensing images have found widespread application in land planning, natural disaster risk assessment, and environmental monitoring [4]. The advantages of satellite remote sensing, such as wide coverage and fewer environmental restrictions, allow for large-scale and operational power line inspection, which greatly enhances the efficiency and pertinence of UAV inspection. Consequently, intelligent power line inspection based on satellite remote sensing has emerged as a crucial research focus.

In comparison to the well-established UAV inspection technology widely used for identifying and researching defects in power line key equipment such as power lines, insulators, power poles, and towers, current studies mostly focus on using satellite remote sensing images for vegetation erosion [5], disaster monitoring [6], [7], and other tasks in the power corridor, with limited involvement in identifying power poles, towers, and other targets. As the spatial resolution of satellite remote sensing images continues to improve and the revisit cycle shortens, researchers have discovered the potential to detect and identify large power infrastructure, such as power towers and substations, from submeter-level satellite remote sensing images. Power towers, as essential infrastructure supporting power lines to ensure their safe operation and maintain a certain distance from the ground, play a critical role in the power grid. Obtaining the position information of power towers from satellite remote sensing images can assist power companies in checking the distribution of power towers, supporting inspection personnel in their work, and aiding in regional power grid planning.

Traditional optical satellite remote sensing image target detection mainly relies on machine learning and image processing techniques, including region selection, feature extraction, and classifier. Feature extraction is a critical step in target detection, as it involves analyzing the image's gray values, textures, and land object spectra, among other information. This information plays a key role in detecting targets in remote sensing images. Currently, the application of machine learning algorithms in power line inspection based on satellite

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Chaojun Shi, Zhenbing Zhao, and Ke Zhang are with the Department of Electronics and Communication Engineering and Hebei Key Laboratory of Power Internet of Things Technology, North China Electric Power University, Baoding, Hebei 071003, China (e-mail: scj@ncepu.edu.cn; zhaozhenbing@ncepu.edu.cn; zhangkeit@ncepu.edu.cn).

Xian Zheng, Zibo Su, and Qiaochu Lu are with the Department of Electronics and Communication Engineering, North China Electric Power University, Baoding, Hebei 071003, China (e-mail: 220222215007@ncepu.edu.cn; 220222215011@ncepu.edu.cn; 220232215009@ncepu.edu.cn).

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remote sensing images has gained traction. Uehara et al. [8] utilized high-order local autocorrelation image features to extract spatial and spectral relationships from multispectral satellite images. They employed AdaBoost as a classifier for patch images, achieving approximately 90% accuracy and less than 80% recall. Meanwhile, Prakash and Kak [9] utilized active learning technology on multispectral images covering 150 000 km^2 in southeast Australia, obtained by World-View2. They applied supervised pre-labeled data to detect transmission towers, ultimately achieving an 80% accuracy rate and a 50% recall rate for transmission line tower detectors. In addition, Gershenson et al. [10] proposed an automated method to estimate the position of power lines in images using indirect indicators of power infrastructure, such as moderate-resolution imaging spectroradiometer (MOIDS) land cover data and night light data. However, traditional optical satellite remote sensing image target detection algorithms are only applicable to specific environments and backgrounds, have poor generalization ability, and are mostly used for extracting large targets such as transmission towers, with little involvement in distribution towers. Meanwhile, they cannot effectively address the issues of noise, complex backgrounds, and varying target scales in satellite remote sensing images.

With the rapid development of deep learning, researchers have recognized the potential for applying object detection methods based on deep learning to better extract power targets in satellite remote sensing images. The object detection method based on deep learning uses a convolutional neural network (CNN) to represent hierarchical high-dimensional features and automatically extract image features. Mainstream target detection methods are primarily divided into two categories: the two-stage region-based target detection algorithm (e.g., region-CNN (RCNN) [11], Fast R-CNN [12], and Faster R-CNN [13]) and the one-stage detection algorithm (e.g., single shot detection (SSD) [14], Retina-Net [15], and you only look once (YOLO) series [16], [17], [18]). Hu et al. [19] constructed four power tower data subsets under different geographical locations and explored the use of three models (Faster R-CNN, YOLOv2, and RetinaNet) to automatically detect power towers from the electric transmission and distribution infrastructure imagery dataset [20]. Ultimately, RetinaNet performed the best, achieving 47% precision, 60% recall, and 53% $F1$ score. The study also observed that a satellite image resolution of at least 0.3 m was required to detect at least half of the power towers. Furthermore, Fang et al. [21] proposed a sample data augmentation method for high-resolution satellite remote sensing ground imaging features to address the issue of insufficient power pole and tower samples. Haroun et al. [22] utilized the RetinaNet deep learning model to detect the location of transmission towers from satellite images, achieving an average accuracy mAP0.5 of 72.45% when the intersection over union (IoU) was 0.5. In addition, they developed a routing algorithm to extract the power corridor area by creating a virtual path between each adjacent detected transmission tower. Currently, most satellite remote sensing image target detection algorithms for power tower based on deep learning rely on the advantages of typical target detect-ion models but have not effectively solved the problems of multiscale power poles and complex backgrounds in satellite remote sensing images.

Optical satellite remote sensing image features can generally be divided into high-level features and low-level features. High-level features refer to abstract semantic information, while low-level features include detailed information such as the spectrum and texture of the image. In CNN architectures, the distribution of features at each layer is related to the scale of the target. In satellite remote sensing images of power towers, distribution towers and transmission towers vary in scale and the difference is significant, making it difficult to simultaneously transfer the features of both to deeper network layers. Moreover, deep learning-based target detection models use the last layer of features extracted by the CNN for classification and localization, which reduces the amount of small target information in the final network output, leading to lower detection accuracy for small targets such as distribution towers. Therefore, utilizing multiscale feature maps and designing multiscale feature fusion modules is particularly crucial. Hou et al. [23] designed a feature fusion strategy through cascading to integrate high-level semantic information and low-level detail information, in order to reduce the loss of small target information during the transmission of CNN networks, thereby enhancing multiscale feature representation. Fu et al. [24] proposed a feature fusion architecture to generate multiscale feature hierarchies, which includes both top–down and bottom–up approaches to fuse shallow and deep features. Zhu et al. [25] introduced the transformer prediction head (TPH)-YOLO model, which uses Transformers to improve the prediction network of YOLOv5, combining self-attention mechanisms for effective detection of multiscale targets. Zhou et al. [26] proposed an attention multihop graph and multiscale convolution fusion network, which includes two subnetworks: a multiscale full CNN and a multihop graph convolutional network (GCN), used for extracting multilevel information from remote sensing images.

The detection of power tower targets in satellite remote sensing images is also challenged by the interference of complex backgrounds. Power towers are highly susceptible to the influence of various similar ground objects in terms of color, shape, and other features, leading to false detection. Given the limited satellite remote sensing data of power towers, achieving target detection in complex backgrounds can be realized by extracting contextual information of the targets, introducing attention mechanisms, or eliminating differences between remote sensing images of various backgrounds through feature transfer and feature embedding. This reduces the uncertainty of target features and enables accurate image interpretation. Hong et al. [27] proposed a high-resolution domain adaptive network (HighDAN), which captures multiscale image representations from parallel high-to-low-resolution subnetworks, generating repeated information between different resolutions efficiently and bridging the gap of remote sensing images from different backgrounds through adversarial learning, effectively reducing interference from the remote sensing image background. To better handle remote sensing images from different sources, Hong et al. [28] later designed a universal remote sensing foundational model, SpectralGPT, which

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adapts to input images of varying sizes, resolutions, temporal sequences, and regions through progressive training, capturing spectral sequential patterns through multitarget reconstruction, and fully utilizing remote sensing data in different scenarios. Zhang et al. [29] proposed a context-aware detection network to eliminate interference from complex backgrounds, which utilizes attention-modulated features and global and local contexts to adapt to different environmental changes of the target. Wang et al. [30] proposed a representation-enhanced state replay network, which jointly optimizes parameters across different branches, enhancing the interaction and fusion of information between different remote sensing images.

In summary, multiscale detection and interference from complex backgrounds hinder the development of target detection algorithms for power tower images in satellite remote sensing. Most existing methods for power tower detection still employ basic one-stage or two-stage main-stream target detection models, which are less effective for small target detection and have not fully addressed issues of multiscale feature fusion and model generalization in complex backgrounds. Therefore, starting from the existing problems in satellite remote sensing detection of power towers, this article aims to provide a target detection framework that can accurately locate power tower positions in satellite remote sensing images under complex backgrounds and effectively distinguish between transmission towers and distribution towers, improving the multiscale detection accuracy of power towers. Leveraging the electric transmission and distribution infrastructure imagery dataset [20], this article constructs a multiscene high-resolution satellite remote sensing power tower dataset (SRSPTD), containing location and category information of power towers. Utilizing the YOLOv5s algorithm, this article proposes the large selective kernel feature fusion YOLO (LSKF-YOLO) network for detecting power towers in high-resolution satellite remote sensing images. The network primarily comprises a large spatial selective kernel attention mechanism fusion module and a multiscale feature information alignment fusion module.

The main contributions of this article are summarized as follows.

1) Utilizing the electric transmission and distribution infrastructure imagery dataset, we selected images from diverse backgrounds and population density regions to create the SRSPTD. The selected images were then cropped and relabeled as transmission tower and distribution tower.
2) To effectively leverage the contextual information of satellite remote sensing images and extend the model's sensitivity field, this article introduces the large spatial selective kernel mechanism (LSKM) in the baseline backbone network. In addition, it utilizes the attention feature fusion module (AFF) to enhance its residual connection, thereby achieving precise positioning of power poles and towers.
3) To address the issue of deep semantic features in the original network's feature pyramid network (FPN) being used more extensively than shallow semantic features, this article incorporates a multiscale feature alignment fusion (MFAF) module in the neck network of the benchmark model. This enables the effective utilization of low-level feature information and facilitates multiscale feature fusion for power towers. Furthermore, to enhance the detection accuracy of the micro tower, the stride convolution in the last two layers of the neck network is replaced by space-to-depth convolution (SPD-Conv) and nonstride convolution. Minimum point distance-IoU (MPDIoU) is used to improve complete-IoU.
4) We evaluate the performance of the proposed LSKF-YOLO on the SRSPTD dataset constructed in this article. The experimental results demonstrate that LSKF-YOLO significantly enhances the accuracy of power tower detection in satellite remote sensing images. In comparison to prior research outcomes, this study facilitates the development of a power tower detection model based on satellite remote sensing images, enabling the automatic detection and classification of transmission and distribution towers by inspectors.

The rest of this article is organized as follows. The related work is described in Section II. Section III introduces the proposed power tower detection method based on satellite remote sensing image and describes the structure and principle of the improved module in detail. Section IV describes the experimental process, including the creation and processing of dataset, and the setting of experimental environment and parameters, and gives the analysis results. Section V draws the conclusion. Data acquisition addresses are described in Data Availability.

II. Related Works

A. Power Lines Inspection Using Satellite Remote Sensing

The inspection of power lines based on satellite remote sensing images primarily involves optical remote sensing satellites and synthetic aperture radar (SAR) data sources [30]. These data sources can be categorized into high-, medium-, and low-resolution images based on their spatial resolutions, with high-resolution images offering greater accuracy and wider applicability. Optical remote sensing satellites, such as the Gaofen-2 satellite [31] and World-View4 satellite [32], rely on sunlight to illuminate the Earth's surface, capturing the reflected light to passively acquire clear, easy-to-interpret images. However, these images are hindered by cloud cover, rain, fog, and nighttime conditions. In power line inspection, optical remote sensing satellites are commonly employed to monitor vegetation encroachment in power corridors. For instance, Mikhalap et al. [33] utilized the normalized difference vegetation index (NDVI) threshold to detect vegetation encroachment in the overhead power line corridors spanning approximately 550 km in the Pskov region of Russia. Their analysis using a geographic information system (GIS) revealed that 84% of the power line corridors required management. Haroun et al. [34] reviewed vegetation detection methods in power transmission corridors using optical remote sensing satellite images, classifying them into three categories: vegetation index-based, stereo matching technology-based, and object detection method-based approaches.

In contrast to optical remote sensing satellites, SAR satellites have a later maturation and operate as an active remote sensing imaging mode. They emit electromagnetic

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waves to the ground multiple times, capturing the reflected signals to indirectly measure the characteristics of the observed objects. TerraSAR-X [35], COSMO-SkyMed [36], and other SAR satellites offer the advantage of all-weather and all-day imaging capabilities, as well as a certain level of surface penetration, addressing the limitations of optical and infrared remote sensing. However, traditional SAR satellites typically have long orbit periods measured in weeks and low data update frequencies, making them suitable for long-term surveillance. SAR satellites are particularly advantageous for large-scale and remote area monitoring. Many studies based on SAR imagery are dedicated to mapping power lines and towers or monitoring hazards around power corridors. For instance, Yan et al. [37] analyzed the backscattering of towers and lines in X-band SAR images from TerraSAR and COSMO-SkyMed satellites, establishing the correlation between the orientation of power lines and their visibility. Sha et al. [38] conducted a time series study using 20 TerraSAR images to examine the backscattering behavior of a power line segment with six wires. In addition, Jingnan [39] assessed the scattering characteristics of icing on transmission towers, demonstrating the potential of high-resolution SAR images in monitoring deformation changes caused by icing under varying weather conditions. Furthermore, Sheng et al. [40] utilized multisource SAR satellite data (X-/C-/L-bands) to monitor geological hazards such as landslides along high-voltage cross-river lines in Hubei province. Despite the widespread application of satellite remote sensing images in power line inspection, most studies remain focused on vegetation erosion and disaster monitoring, with few addressing the identification and detection of power infrastructure in the images. Thus, there is a need for research on the development of detection and recognition algorithms for power poles and towers based on satellite remote sensing images.

B. Object Detection Based on Satellite Remote Sensing Image

Compared to images obtained through UAV inspection, satellite remote sensing images exhibit varying target scales, uneven distribution, and an abundance of small targets. At the same resolution, these images encompass targets of different scales and types, with nonuniform distribution. Consequently, detecting targets of various sizes often presents a multiscale detection challenge. The conventional approach involves utilizing an FPN [41] to adjust the image to multiple scales and employing a top–down structure to merge these features. However, FPN lacks the capability to convey positioning information from the low-level feature map to the high-level feature map. Furthermore, the feature transfer between layers relies on adjacent layers, resulting in imbalanced feature fusion. Subsequently, path aggregation network (PANet) [42] and bidirectional feature pyramid network (BiFPN) [43] were introduced to enhance the feature information flow of FPN. PANet introduced a bottom–up path structure and utilized shallow network features to integrate FPN features, while BiFPN incorporated context and weight information based on PANet to achieve a more balanced integration of features at different scales, thereby obtaining richer semantic information. Semantic flow for fast and accurate scene parsing (SFNet) [44] aligns different levels of features with semantic streams to enhance FPN performance within the model. Nevertheless, existing FPN-based structures exhibit an excessive number of pathways and indirect interaction modes, leading to potential cross-layer information loss and difficulties in meeting the requirements of satellite remote sensing power tower detection in practical scenarios. U-Net in U-Net (UIU-Net) [45] introduced the resolution maintenance depth monitoring module (RM-DS) and interactive cross-attention module (IC-A). These modules utilize multiple intermediate layers to acquire comprehensive and distinguishable features, enabling the learning of depth multiscale features for enhancing global context representation and addressing the tradeoff between feature resolution and network depth. Gold-YOLO [46] proposed an information collection and distribution mechanism, where the feature alignment fusion injection module selectively fuses low-level or high-level feature information to ensure information integrity.

In the context of satellite remote sensing transmission and distribution tower detection, accurately representing points of interest is particularly crucial due to the smaller size of the target compared to the image background. One widely utilized method for this purpose is the attention mechanism. This mechanism, which simulates the distribution of human visual and auditory attention, assists the model in assigning varying weights to different parts of the input, thereby extracting more critical information. Spatial attention modules, such as gather-excite network (GE-Net) [47] and spatial group-wise enhance (SGE) [48], enhance specific target areas while diminishing irrelevant background areas by generating weight masks for each location and adjusting the output accordingly. Meanwhile, channel attention modules, such as squeeze and excitation network (SE-Net) [49] and efficient channel attention network (ECA-Net) [50], automatically determine the importance of each feature learning. These modules use global average information to reweight feature channels, emphasizing important features and suppressing nonessential ones. Convolutional block attention module (CBMA) [51] integrates spatial and channel domain attention to create a more comprehensive attention method. In addition, self-attention mechanisms, such as dual attention network (DA-Net) [52] and criss cross network (CC-Net) [53], as variants of the attention mechanism, leverage inherent information within features to maximize attention interaction, thereby reducing reliance on external information.

Transformer-based models [54], such as vision transformer (ViT) [55], [56] and Swin-transformer [57], [58], have emerged as competitive alternatives to CNNs. The key to their success lies in their large receptive fields (RFs), which provide a similar acceptance domain to the self-attention mechanism. It is worth noting that a suitably large convolutional kernel network can achieve performance comparable to the transformer network. ConvNeXt [59], RepLKNet [60], and Conv2Former [61] have all demonstrated the significant role of large kernel convolutions in modulating convolution features within richer contexts. Attention mechanisms based on kernel selection are also worthy of consideration. Selective kernel network (SKNet) [62] uses multiple branches with different kernel sizes and selectively combines them along

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the channel dimension. Similarly, self-calibrated network (SCNet) [63] employs branch attention to capture richer information and spatial attention to enhance localization. Large selective kernel network (LSK-Net) [64] adopts the concept of SKNet but adaptively aggregates macrokernel information across spatial dimensions, effectively utilizing contextual information in images.

III. Proposed Methods

Considering YOLO continued status as the state-of-the-art (SOTA) model with a balance between accuracy and speed in the domain of small models, the YOLOv5 [65] algorithm stands out for its small model size and excellent speed in object detection. YOLOv5s was selected as the baseline for this study. The input consists of a three-channel RGB image with a feature size of 640 × 640 × 3. The backbone is CSPDarknet53 [66], known for its robust feature extraction and computational efficiency, mainly composed of Conv, C3, spatial pyramid pooling (SPPF), and bottleneck modules. YOLOv5s adopts the pyramid structure of PANet in the neck, using top–down horizontal connections to construct high-level semantic features at various scales and adding bottom–up structures to reinforce positioning information. The output section includes three Yolo head detectors that can output three different dimensional feature maps. YOLOv5s utilizes the Focal_loss function for target and classification losses, and the CIoU loss function for positioning losses.

This study addresses challenges related to differences in the resolution of satellite remote sensing power tower images, difficulty in extracting features of small distribution tower targets, complex image backgrounds, and significant size differences between transmission and distribution towers. We improved and optimized the YOLOv5s network, resulting in a network suitable for power pole detection and recognition in satellite remote sensing images, named LSKF-YOLO. As illustrated in Fig. 1, we introduced three new contributions to the LSKF-YOLO network architecture. We incorporated the attention feature fusing-large spatial selective kernel mechanism (AFF-LSKM) into the backbone network to enhance its feature extraction capability and expand the RF, thus better utilizing context information. In addition, we employed the space-to-depth MFAF (SPD-MFAF) module in the neck network to fuse feature information, improving the sensitivity of the network to small targets through the space-to-depth convolution layer while retaining low-level feature information. Furthermore, we introduced MPDIoU_Loss to optimize CIoU_Loss for better position positioning and category recognition of power towers. The LSKF-YOLO network structure is shown in Fig. 2.

A. Attention Feature Fusion Large Selective Kernel Mechanism Module

Satellite remote sensing images possess unique features, making them particularly well suited for remote sensing missions. The concept of large kernel attention provides a sensory field akin to the self-attention mechanism. Through the use of deep convolution, deep dilation convolution, and 1 × 1 convolution, large convolutional kernels can be constructed with fewer parameters and computations. Inspired by LSK-Net [64], this article proposes the AFF-LSKM module, as depicted in Fig. 3. It primarily consists of a large kernel spatial selection subblock, a feedforward neural network complex subblock [multilayer perceptron (MLP)], and an AFF [67].

The large kernel spatial selection subblock encompasses the LSK module (as depicted in Fig. 4), which consists of the large kernel convolution sequence and the spatial kernel selection mechanism.

Large convolutional sequences are constructed through a hierarchical decomposition into deep convolutional sequences with large kernel sizes and increasing dilation rates. Here, we define the convolution kernel size as K, expansion rate as d, and RF. The initial expansion rate is set to 1, with an upper limit to prevent expansion convolution from introducing gaps between feature maps.

Both the initial RF and the convolution kernel have the same initial size, and their relationship can be expressed as follows:

$$\mathrm{RF}_i = d_i(K_{i-1} - 1) + \mathrm{RF}_{i-1} \tag{1}$$

$$K_{i-1} \le K_i, d_{i-1} < d_i \le \mathrm{RF}_{i-1}. \tag{2}$$

The formula indicates that the RF expands with an increase in both kernel size and expansion rate. To describe the contextual information features of input feature X across different ranges, a series of deep convolutions with varying RFs are used for decomposition. The decomposition nuclei are expressed as $D_i(D_0 = X)$. Assuming that there are N decomposition nuclei, dw_i represents a deep convolution with kernel K_i and expansion rate d_i. After decomposition, each kernel is processed by a 1 × 1 convolution layer $\mathrm{Cl}_{1\times1}$. This decomposition process is illustrated in the following equation:

$$D_{i+1} = dw_i(D_i), \tilde{D}_i = \mathrm{Cl}_{1\times1}(D_i), \quad \text{for } i \text{ in}[1, N]. \tag{3}$$

To enhance the network's ability to focus on the most relevant spatial context region for improved target localization, we introduce a spatial selection mechanism to extract feature maps from large convolution kernels of different scales. Initially, we connect the features obtained from different nuclei with varying RF ranges. Subsequently, we apply channel-based mean pooling P_{avg} and maximum pooling $P_{\max}$ to $\tilde{D}$ in order to effectively extract spatial relationships and derive the spatial attention graph $S\tilde{A}$, Each spatial attention graph is then interconnected to facilitate information interaction between them, and channel 2 is transformed into channel N using $\mathrm{Cl}^{2\to N}$ [see (4)]. Next, a separate spatial selection mask $S\widehat{A}_i$ [see (5)] for each decomposition megacolor is obtained through the Sigmoid activation function. Subsequently, the features in the decomposition kernel are appropriately weighted and fused through the convolution layer to obtain the feature S of interest [see (6)]. Finally, the ultimate output is the fusion of the input feature map and the features of interest [see (7)]. The entire process can be represented as follows:

$$S\tilde{A} = \mathrm{Cl}^{2\to N}\left(\left[P_{\mathrm{avg}}\left(\tilde{D}\right); P_{\max}\left(\tilde{D}\right)\right]\right) \tag{4}$$

$$S\widehat{A}_i = \sigma\left(S\tilde{A}_i\right) \tag{5}$$

$$S = CL\left(\sum_{i=1}^{N} S\widehat{A}\cdot\tilde{D}_i\right) \tag{6}$$

$$Y = X \cdot S. \tag{7}$$

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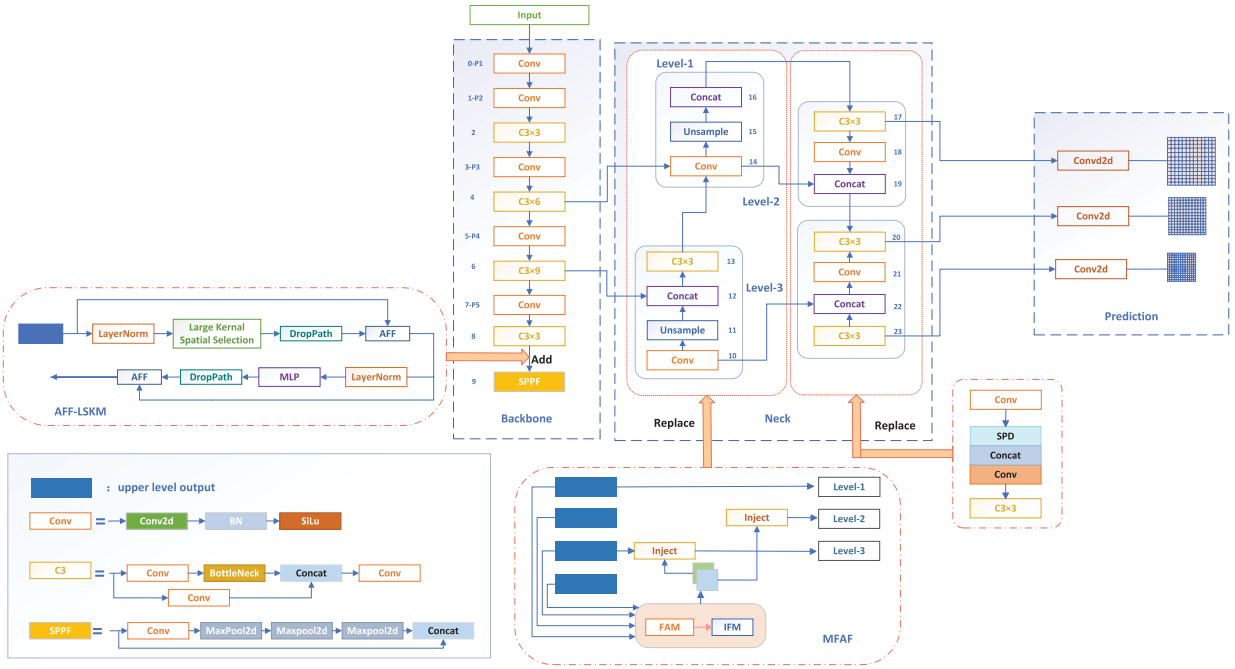


Fig. 1. Improved technology used in LSKF-YOLO. A large selective kernel mechanism LSKM is added to the original backbone network, and the feature attention fusion module AFF is used to improve the residual connection and enhance the feature extraction capability of the backbone. The MFAF structure MFAF and SPD-Conv are introduced to replace the original neck network to improve the multiscale and small target detection capability of the model.

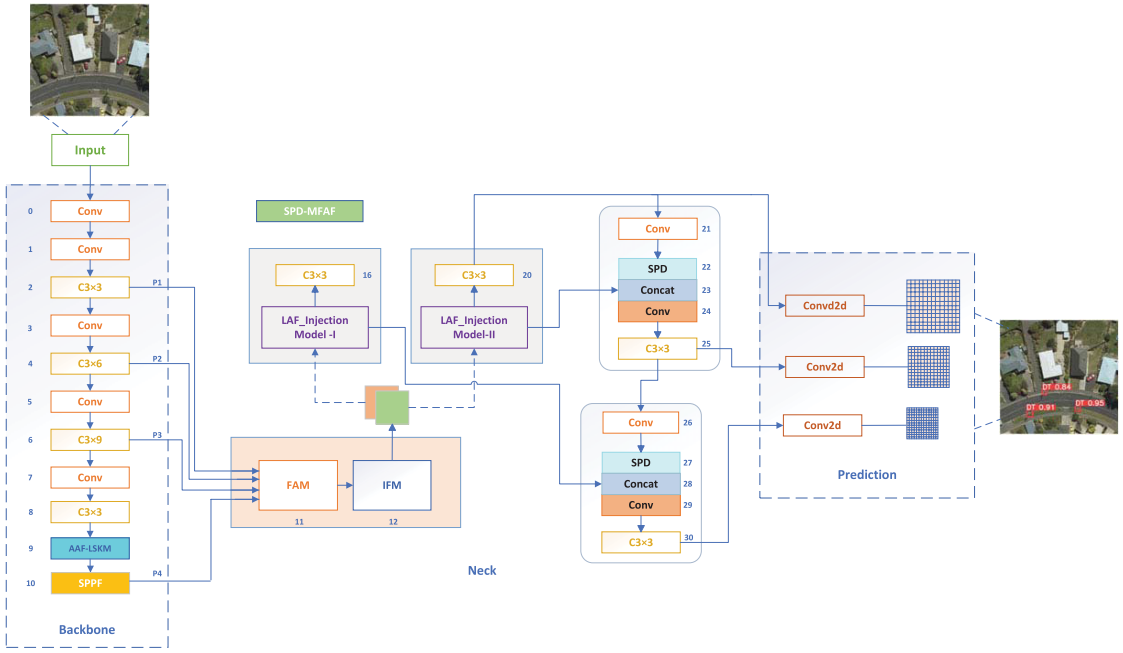


Fig. 2. LSKF-YOLO structure.

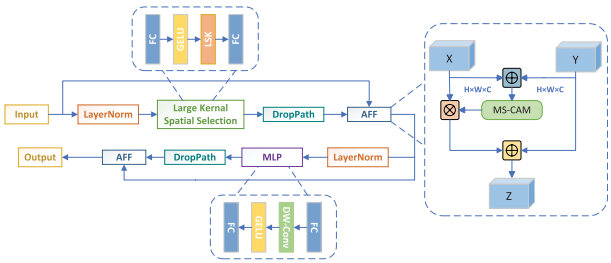


Fig. 3. AFF-LSKM structure.

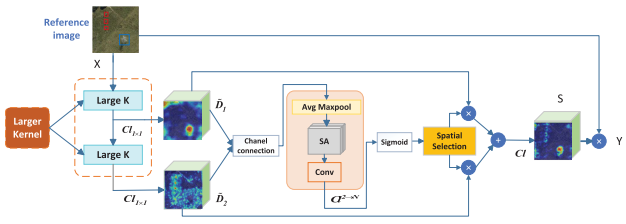


Fig. 4. Schematic of LSK subblocks.

To address the issues of gradient vanishing and weight matrix degradation resulting from an increase in the number of network layers, LSKM incorporates residual connections between each sublayer and replaces the feature fusion module at the connection with the AFF module. The AFF module

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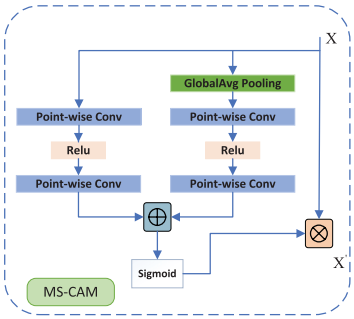


Fig. 5. MS-CAM module in AFF.

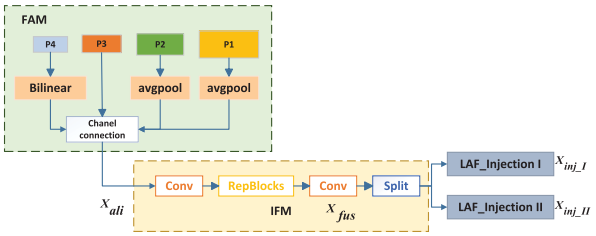


Fig. 6. FAM and IFM module structure.

alters the distribution of feature weights by adjusting the attention weights across different scales of channel dimensions. In this process, the input feature graph serves as X for the AFF module, the output feature graph of DropPath as Y, and the resulting output feature graph of AFF as Z. This effectively enhances the focus on features within the channel and spatial dimensions. This principle can be expressed by the following equation:

$$Z = M(X \oplus Y) \otimes X + (1 - M(X \oplus Y) \otimes Y. \quad (8)$$

The core multiscale channel attention mechanism (MS-CAM) (Fig. 5) emphasizes channel scaling through point-by-point convolution, concurrently extracting local and global context features within the channel attention module.

AFF-LSKM augments the model's RF, capturing diverse local information and strengthening the backbone network's capability to extract targets within complex backgrounds.

B. Space-to-Depth Mutiscale Feature Alignment Fusion Network

The neck network of the benchmark model adopts PANet to better integrate low-level semantic information and prevent information loss caused by indirect interaction in the network. This article introduces the MFAF module to replace the first two modules of the original neck network in the first two layers of structure, effectively utilizing the low-level semantic features. Furthermore, in order to enhance the detection ability of small targets such as distribution towers, this article utilizes the SPD-conv to replace the original step convolution layer in the last two neck networks. This prevents the fuzzy feature map caused by the increase of network depth from affecting the detection effect of the detection head. The network structure is named SPD-MFAF.

The MFAF module consists of a feature alignment module (FAM), an information fusion module (IFM) (shown in Fig. 6), and a lightweight adjacent layer fusion (LAF)_injection module (shown in Fig. 7). In the FAM, the output features $P1$, $P2$, $P3$, and $P4$ of the backbone network in Fig. 2 are selected for alignment fusion to obtain high-resolution features that retain small target information, resulting in the feature X_{ali}. An AvgPool operation is used to downsample the input features and achieve a uniform size. The smallest feature, $P3$, of the output of the backbone network in Fig. 2 is selected as the alignment feature to retain more underlying information, resulting in the feature X_{ali}.

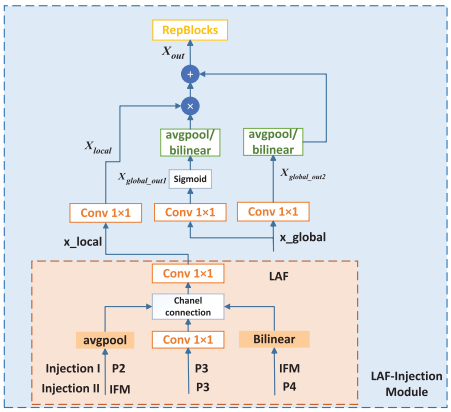


Fig. 7. LAF_injection module.

The IFM comprises multiple layers of reparameterized convolution blocks (Repblocks) and split operations. The RepBlock takes the output feature of FAM as input and produces the fusion feature X_fus. The features generated by RepBlock are then split into input features $X_{\text{inj_I}}$, $X_{\text{inj_II}}$ of LAF_injection_module I and LAF_injection_module II in the channel dimension, before being fused with features of different levels, as illustrated in the following equations:

$$X_{\text{ali}} = \text{FAM}([P1, P2, P3, P4]) \quad (9)$$

$$X_{\text{fus}} = \text{RepBlock}(X_{\text{ali}}) \quad (10)$$

$$X_{\text{inj_I}}, X_{\text{inj_II}} = \text{Split}(X_{\text{fus}}). \quad (11)$$

In the LAF_injection_module, the attention operation is employed to fuse information, incorporating two components: local feature x_local and global information x_global. The local features of LAF_injection_module I are derived from $P2$, $P3$, and IFM information fused using the LAF module. Similarly, LAF_injection_module II provides local features from IFM, $P3$, and $P4$ information integrated in the LAF module, while the global features are derived from the IFM output information. The computation of x_global involves two different convolution layers to capture the characteristics of the two branches, whereas x_local is computed using only one convolution layer. Subsequently, these three output features are carefully fused through calculations. Due to the distinctions between local and global features, the output features are scaled based on the size of the local information using average pooling or bilinear interpolation to maintain alignment. Following the completion of fusion, RepBlock is added to further extract the fusion information, as depicted in the following equations:

$$X_{\text{global_out1}} = \text{resize}(\text{sigmoid}(\text{Conv}(x_\text{global})) \quad (12)$$

$$X_{\text{global_out2}} = \text{resize}(\text{Conv}(x_\text{global})) \quad (13)$$

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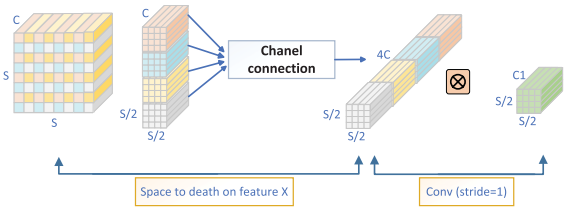


Fig. 8. Schematic of SPD-Conv.

$$X_{\text{attn}} = \text{Conv}(x_\text{local}) * X_{\text{global_out1}} + X_{\text{global_out2}} \quad (14)$$

$$X_{\text{out}} = \text{RepBlock}(X_{\text{attn}}). \quad (15)$$

To enhance the model's capability in detecting small targets such as distribution towers, this study introduces the replacement of the step convolution with space to deep convolution layer and nonstep convolution in the last two layers of the neck network. The SPD convolution downsamples the feature mapping. For an input feature X with a scale of $S \times S$ and a channel dimension of C, subfeature mappings are derived, with the scale factor set at 2. This process yields four subfeature maps, each halving the scale to $S/2$ while maintaining the channel dimension. These feature maps are connected in the channel dimension, resulting in a channel dimension of $4C$. Finally, a nonstep convolution with a step size of 1 is applied to obtain the output feature map. This approach effectively enhances the model's sensitivity to small targets. The schematic illustrating this process is shown in Fig. 8.

C. MPDIoU

In target detection, the boundary box regression loss function plays a critical role in quantifying the disparity between the actual frame and the predicted frame. In this research, we introduce MPDIoU_Loss as an enhancement to CIoU_Loss. MPDIoU is a novel bounding box similarity comparison metric that relies on the minimum point distance. It encompasses all factors considered in existing loss functions, encompassing overlapping and nonoverlapping regions, center point distance, width, and height deviation while streamlining the calculation process. Representing the real box as A and the predicted box as B, the traditional IoU can be expressed as follows:

$$\text{IoU} = \frac{A \cap B}{A \cup B}. \quad (16)$$

The CIoU adopted in the baseline model comprehensively considers the center point distance and aspect ratio, and its publicity is as follows:

$$\text{CIoU} = \text{IoU} - \frac{\rho^2(A, B)}{C^2} - \alpha V \quad (17)$$

$$V = \frac{4}{\pi^2}\left(\arctan\frac{w^A}{h^A} - \arctan\frac{w^B}{h^B}\right)^2 \quad (18)$$

$$L_{\text{CIoU}} = 1 - \text{CIoU} \quad (19)$$

where $\rho^2(A, B)$ denotes the Euclidean distance between the center point of the prediction box and the center point of the real box, while C^2 represents the diagonal length of the smallest enclosing rectangle. In addition, w^A and h^A represent the width and height of the real box, respectively, while w^B and h^B represent the width and height of the prediction box, respectively. The term L_{CIoU} denotes the loss of CIoU.

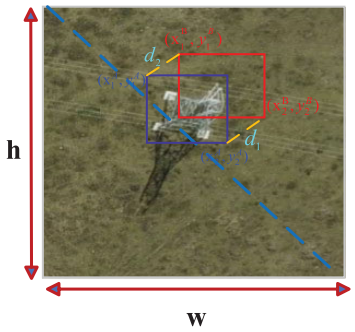


Fig. 9. Schematic of MPDIoU.

To address the issue where the regression loss function cannot be optimized when the predicted frame and the real frame have the same aspect ratio but significantly different widths and heights, we draw inspiration from the geometric properties of bounding boxes. MPDIoU forcibly approximates each bounding box predicted by the model to its real box by minimizing the loss function during the training phase. The four point coordinates of the boundary box are utilized to encompass all the factors of the existing boundary box regression loss function. Its schematic is illustrated in Fig. 9.

Assume that the width and height of the input image are denoted as w and h, and the top-left and bottom-right coordinates of the real box and the prediction box are expressed as (x_1^A, y_1^A), (x_2^A, y_2^A), (x_1^B, y_1^B), and (x_2^B, y_2^B). The mathematical expression for the distance between the top-left corner and the bottom-right corner of the two boundary boxes is as follows:

$$d_1^2 = \left(x_1^B - x_1^A\right)^2 + \left(y_1^B - y_1^A\right)^2 \quad (20)$$

$$d_2^2 = \left(x_2^B - x_2^A\right)^2 + \left(y_2^B - y_2^A\right)^2 \quad (21)$$

where d_1 represents the distance from the top-left corner and d_2 represents the distance from the bottom-right corner, and then, MPDIoU and MPDIoU_Loss can be expressed as

$$\text{MPDIoU} = \frac{d_1^2}{w^2 + h^2} - \frac{d_2^2}{w^2 + h^2} \quad (22)$$

$$L_{\text{MPDIoU}} = 1 - \text{MPDIoU}. \quad (23)$$

Therefore, all components of the current boundary box regression loss function can be determined by the four point coordinates. The conversion formula is as follows:

$$|C| = \left(\max\left(x_2^A, x_2^B\right) - \min\left(x_1^A, x_1^B\right)\right) * \left(\max\left(y_2^A, y_2^B\right) - \min\left(y_1^A, y_1^B\right)\right) \quad (24)$$

$$x_c^A = \frac{x_1^A + x_2^A}{2}, y_c^A = \frac{y_1^A + y_2^A}{2}, y_c^B = \frac{y_1^B + y_2^B}{2}, x_c^B = \frac{x_1^B + x_2^B}{2} \quad (25)$$

$$w_A = x_2^A - x_1^A, h_A = y_2^A - y_1^A, w_B = x_2^B - x_1^B, h_B = y_2^B - y_1^B \quad (26)$$

where $|C|$ represents the smallest enclosing rectangular area covering A and B and (x_c^A, y_c^A) and (x_c^B, y_c^B) represent the

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coordinates of the center point of the real box and the prediction box, respectively. In addition, w^A and h^A represent the width and height of the real box, respectively, and w^B and h^B represent the width and height of the predicted box, respectively.

Upon evaluating (24)–(26), we can deduce that all factors considered in the current loss function can be determined by the coordinates of the top-left and bottom-right points, indicating that MPDIoU is not only comprehensive but also simplifies the calculation process.

IV. Experimental Results and Analysis

In this section, we will verify the performance of the proposed method using the constructed power tower SRSPTD. We will begin by describing the dataset used, followed by introducing the experimental environment and parameters, as well as the selection of evaluation indicators. The subsequent step involves presenting the evaluation results, including visualizing the experimental process and conducting ablation experiments to analyze the influence of the module. Finally, we will compare the proposed method with other CNN-based power tower detection methods to demonstrate the latest progress of our approach.

A. Dataset and Training

The images utilized in this article are sourced from the electric transmission and distribution infrastructure image dataset compiled by Duke University [20]. This dataset comprises fully annotated high-resolution satellite and aerial images of transmission and distribution infrastructure, encompassing an area of approximately 321 km^2 across 14 cities in six countries on five continents.

It covers diverse terrain types and human settlement densities. Specifically, this article focuses on images from four states in the United States (Tucson, AZ; Hartford, CT; Colwich and Maize, KS; and Wilmington, NC), as well as images from Tauranga and Dunedin, New Zealand, obtained from Linz, Austria, which form a subset of the data. The images from these six selected regions represent four types of terrain: desert, plain, forest, and coastal, and three types of human habitation density areas: suburban, rural, and urban. The details of the data subsets are presented in Table I.

The original images in the dataset had resolutions ranging from 3800 to 12 000 pixels, and some had an aspect ratio that made direct training impractical. To better suit the model's training requirements, all subset images were cropped from top to bottom and left to right to 512 × 512 images. In addition, since the original dataset's annotations were in polygon format, which was not suitable for the target detection model, we used labelImg to reannotate the images, categorizing annotations into transmission towers and distribution towers. As the distribution towers were challenging to observe in satellite images, their own shadows were labeled as a unit with themselves during the annotation process to provide more information to the model. An example of the power tower SRSPTD is shown in Fig. 10. Images with empty or out-of-range labeling targets due to cropping were removed. To address the category imbalance between transmission towers and distribution towers, the experimental data were divided into training and verification sets in a 4:1 ratio using fivefold cross validation, resulting in 2192 training set images and 548 verification set images. To enhance the robustness of the model in dealing with images from various environments, we have performed data augmentation on our experimental data. For geometric distortions, we have applied random cropping, translation, and rotation to the data. For photometric distortions, we have adjusted the hue, saturation, and value of the images. In addition, we have employed the Mixup and Mosaic methods to augment multiple images. The Mixup method involves randomly selecting two samples from the training images and generating new training samples through a weighted sum based on random coefficients. This approach helps the model learn more linear relationships between classes and improves its generalization ability. The Mosaic method, on the other hand, creates a new image by combining four different images, enriching the background of the detected object. This not only increases the diversity of the training data but also helps the model learn to handle occlusions and different backgrounds. By using these data augmentation techniques, the model is exposed to a variety of data transformations during training, which is crucial for enhancing the model's performance in real-world applications.

The loss function of the LSKF-YOLO model includes classification loss, confidence loss, and positioning loss. The classification loss adopts Focal_loss from RetinaNet, allowing for the adjustment of weights for challenging samples during training and focusing the center of gravity on such samples. In addition, the confidence loss is managed by binary crossentropy loss (BCE_Loss), while the regression loss is handled by MPDIoU. To enhance the model's focus on negative samples during training, a slide weighting loss is introduced in the experiment [68]. The average MPDIoU of all bounding boxes obtained is utilized as the threshold α, with values smaller than the threshold considered as negative samples and those larger as positive samples. Subsequently, the weighted function slide is employed to assign higher weights to samples near the threshold. The slide function is expressed as follows:

$$g(x) = \begin{cases} 1x < \alpha - 0.1 \\ e^{1-\alpha}\alpha - 0.1 \le x \le \alpha + 0.1 \\ e^{1-x}x > \alpha + 0.1. \end{cases} \tag{27}$$

The experiment was conducted using the PyTorch 1.13.1 framework, with an NVIDIA RTX 3090 GPU and CUDA 11.7 environment for both training and inference tests. The study utilized the officially published YOLOv5 pretraining weights, and the stochastic gradient descent (SGD) optimizer was employed to iteratively update the network parameters. During training, the initial learning rate for weight was set to 0.001, and the momentum parameter was set to 0.937. All models used in the experiments had a batch size of 16 and were trained for 300 epochs.

B. Evaluation Criteria

In this article, the evaluation of the proposed model's detection and recognition performance for the binary classification problem of transmission towers and distribution towers

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TABLE I
CASE OF DATA SUBSET

Source dataset	Country	City	Area(sq.km)	Width	Height	Resolution (meter)	Rural/ Urban	Terrain
UCOMN	USA	Hartford CT	56.3	10000	10000	0.15†	Suburban	Built environment
USGS (Orthoimagery)	USA	Wilmington, NC	27.9	10000	10000	0.15†	Suburban	Coastal
		Tucson, AZ	39.4	11856*	11936*	0.15†	Urban	Desert
		Colwich KS & Maize, KS	34.8	10000	10000	0.15†	Rural	Plains
LINZ (Orthoimagery)	New Zealand	Tauranga	2.8	3840	5760	0.125	Urban	Forest plains
		Dunedin	2.1	3840	5760	0.30	Rural	plains

†0.5 feet = 0.1524m ~ 0.15m

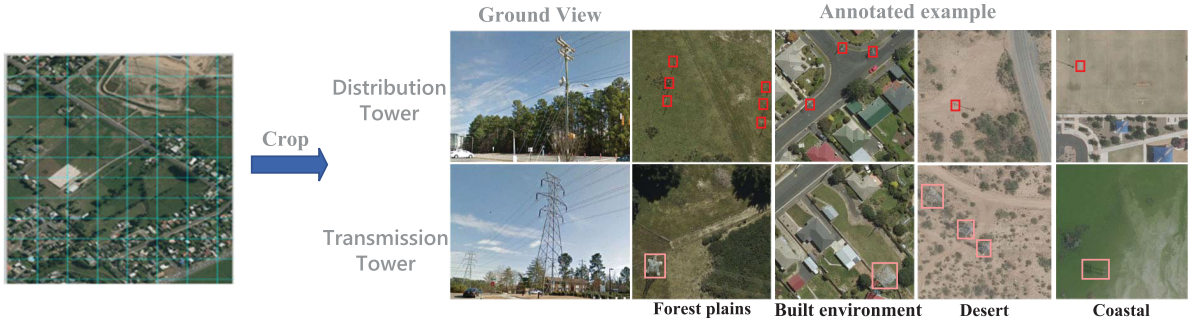


Fig. 10. Sample of annotated images in the SRSPTD dataset.

TABLE II
RESULTS OF ABLATION EXPERIMENTS ON THE SRSPTD DATASET

METHOD				P（%）	R（%）	F1 score	mAP0.5(%)	Params（MB）
YOLOv5s	AFF-LSKM	SPD-MFAF	MPDIoU					
√	×	×	×	77.81	67.92	0.725	73.74	13.77
√	√	×	×	76.02	73.49	0.747	75.93(+2.19)	17.82
√	×	√	×	76.61	70.86	0.736	75.56(+1.82)	16.12
√	×	×	√	78.39	67.83	0.727	74.23(+0.49)	13.67
√	√	√	×	78.63	73.29	0.762	77.13(+3.39)	23.30
√	√	√	√	78.78	73.74	**0.764**	**77.47(+3.74)**	23.28

involves selecting the precision, recall, and average accuracy mAP as evaluation metrics. TP represents the number of correctly detected real power towers, FP represents the number of incorrectly detected power towers, and FN represents the number of undetected power towers.

The precision is defined as the proportion of correctly detected real power towers by the algorithm in all positive predictions, reflecting the model's ability to distinguish negative samples

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}. \tag{28}$$

The recall rate represents the proportion of detected true power tower in all true values, reflecting the model's ability to identify positive samples. Recall is defined as

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \tag{29}$$

The $F1$ score is defined as the harmonic average of precision and recall. This comprehensive consideration helps to mitigate issues caused by category imbalance. The definition of $F1$ score is

$$F1\text{score} = \frac{2 * \text{precision} * \text{recall}}{\text{precision} + \text{recall}} \tag{30}$$

mAP is used to measure the overall effect of the model, and the larger the mAP, the higher the accuracy of the model in

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positioning and recognition, which is defined as follows:

$$\text{mAP} = \frac{\sum_{i=1}^{N} \text{AP}}{N} \tag{31}$$

$$\text{AP} = \int_0^1 P(R)d(r). \tag{32}$$

In addition to performance evaluation indexes, this article also selects network model size (MB) and floating-point operation (FLOP) as efficiency-related evaluation indexes.

C. Ablation Experiment

To evaluate the detection performance of the proposed LSKF-YOLO algorithm, we conducted ablation experiments to evaluate the impact of each algorithm enhancement (large selective kernel attention fusion module, MFAF network, and loss function improvement) on its performance. The performance metrics included precision [$P(\%)$], recall [$R(\%)$], $F1$ score, average precision (mAP0.5), and model size.

Table II presents the ablation results of LSKF-YOLO on the SRSPTD dataset, indicating that the LSKF-YOLO model achieved an increase of 3.9% in precision of $F1$ score and 3.74% in mAP0.5 compared to the original baseline model. The improved LSKF-YOLO model outperformed the original YOLOv5s algorithm.

Both AFF-LSKM and SPD-MFAF networks contributed to enhancing the model performance. When AFF-LSKM was added to the baseline model backbone network alone, model recall significantly improved, while precision slightly decreased. However, $F1$ score and mAP0.5 both showed improvement, increasing by 2.2% and 2.19%, respectively. When the neck network was adjusted to SPD-MFAF alone, the model improved in recall, $F1$ score, and mAP0.5 by 2.94%, 1.1%, and 1.82%, respectively, with a slight decrease in precision by 1.2%. Comprehensive utilization of both improvements led to improved performance across all model indexes, with $F1$ score and mAP0.5 increasing by 3.70% and 3.39%, respectively. In summary, the AFF-LSKM mechanism and the SPD-MFAF network enhanced the model's localization and recognition ability for power towers. In addition, the use of MPDIoU in the model to improve CIoU resulted in a 0.49% improvement in model mAP0.5.

Furthermore, different neck structures on YOLOv5 were studied, including PANet (baseline), BiFPN, MFAF, and SPD-MFAF, to verify their performance. Table III illustrates that the performance of PANet was inferior, while BiFPN and MFAF structures slightly underperformed compared to the SPD-MFAF structures. Considering the model detection ability, the SPD-MFAF structure appeared more suitable for multiscale power tower detection.

In addition, experiments were conducted to investigate the impact of attention mechanisms on the network, including no attention mechanism (baseline) and adding SE and CBAM, LSKM, and AFF-LSKM to find the best-performing attention mechanism. Table IV displays the experimental effects of adding different attention mechanisms to the network backbone, confirming that AFF-LSKM exhibited the best performance.

TABLE III
COMPARISON RESULTS OF DIFFERENT NECK STRUCTURES

MODEL	P	R	F1 score	mAP0.5
Baseline（PANet）	77.81	67.92	0.725	73.74
BiFPN	78.13	68.09	0.727	75.26
MFAF	76.34	70.95	0.735	75.34
SPD-MFAF	76.61	70.86	**0.736**	**75.56**

TABLE IV
COMPARISON RESULTS OF DIFFERENT ATTENTION MECHANISMS IN BACKBONE

MODEL	P	R	F1 score	mAP0.5
Baseline	77.81	67.92	0.725	73.74
SE	76.52	69.85	0.730	74.24
CBMA	75.64	71.95	0.737	75.58
LSKM	77.89	70.18	0.739	75.67
AFF-LKSM	76.02	73.49	**0.747**	**75.93**

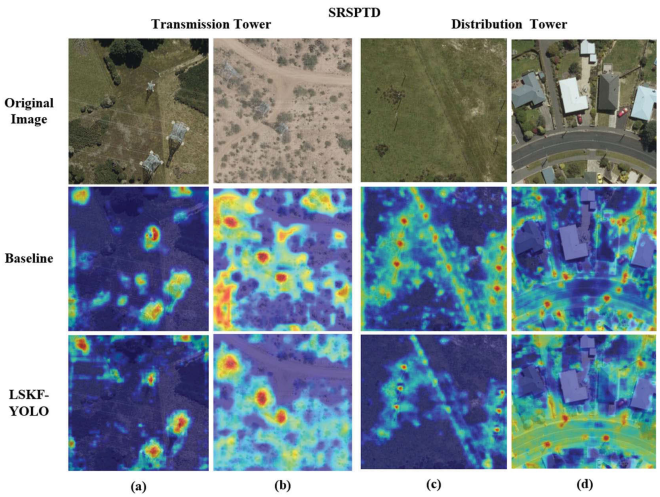


Fig. 11. Heat maps of YOLOv5s and LSKF-YOLO for transmission tower and distribution tower obtained by Grad-CAM. (a) Comparison of heat map of transmission tower in forest plain. (b) Comparison of heat map of transmission tower in desert. (c) Comparison of heat map of distribution tower in forest plain. (d) Comparison of heat map of distribution tower in built environment.

The interpretability of the improved model method was analyzed using Grad-CAM in this article. Fig. 11 presents the heat maps of transmission towers and distribution towers obtained by Grad-CAM for YOLOv5s (baseline) and LSKF-YOLO on the SRSPTD dataset. The results indicated that LSKF-YOLO showed higher attention to the region where the detection target was located. The proposed model demonstrated stronger neglect of nontarget regions in the presence of different backgrounds and better focus on transmission towers of different scales. Moreover, the LSKF-YOLO model has shown promising results in improving the localization ability

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TABLE V
COMPARISON BETWEEN DIFFERENT DETECTION METHODS ON THE SRSPDT DATASET

MODEL	AP（DT）	AP（TT）	mAP0.5	Params（MB）	Flops (G)
Faster- RCNN	53.6	80.9	67.3	28.3	473.2
SSD512	56.4	84.6	70.5	24.6	30.8
RetinaNet	57.3	83.4	72.2	80.2	77.6
YOLOv4	66.2	82.4	74.3	14.8	20.5
YOLOv5-S	65.3	82.2	73.7	13.7	16.2
YOLOv7-Tiny	67.1	82.1	74.6	12.3	15.5
YOLOX-S	66.3	84.5	75.4	21.1	24.8
Deformable-DETR	-	-	74.8	40.2	173.3
YOLOv8-S	65.2	84.2	74.7	11.13	28.6
TPH-YOLO	66.5	83.2	74.9	13.67	16.0
Gold-YOLO	67.8	83.6	75.7	21.51	46.03
YOLO-NAS	66.1	84.3	75.2	19.02	-
LSKF-YOLO	**68.0**	**86.9**	**77.4**	23.2	22.6

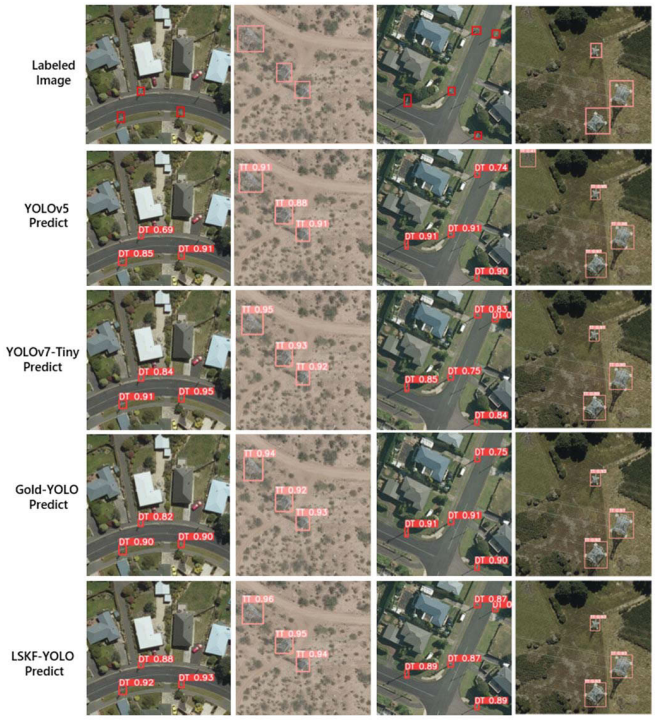


Fig. 12. Sample detection results from YOLOv5-S, YOLOv7-Tiny, Gold-YOLO, and LSKF-YOLO.

for small targets such as distribution towers, particularly in densely populated areas. The model's success in focusing on positive sample areas while reducing attention on irrelevant environmental information demonstrates the effectiveness of the large core space selective attention fusion module and the SPD-MFAF network in enhancing feature representation and detection capabilities for various scale transmission and distribution towers, as well as small targets such as distribution towers. The success of the LSKF-YOLO model highlights the potential of incorporating advanced techniques such as selective attention mechanisms and feature fusion modules in deep learning-based object detection models for satellite imagery analysis.

D. Comparison of Different Target Detection Network

To validate the reliability of the LSKF-YOLO model's performance, we compared its performance with other models on the SRSPTD dataset, as shown in Table IV. Due to the lack of models specifically designed for satellite remote sensing image power pole detection, our experiments mainly compared the LSKF-YOLO model with detection models Faster R-CNN, SSD, and RetinaNet mentioned in related articles [19], [20], where all three models utilized VGG16 as the backbone network. In addition, we use the end-to-end object detection model Deformable-DETR [69], as well as other models in the YOLO family (YOLOv4, YOLOv7-Tiny, YOLOX-S, and YOLOv8-S) as comparison models. At the same time, it is compared with the SOTA target detection models TPH-YOLO [25], Gold-YOLO [46], and YOLO-Neural Architecture Search (NAS) [70] proposed in recent years. The evaluation was based on mAP0.5, params, and Flops as metrics.

As shown in Table V, compared to other detection models, the proposed LSKF-YOLO model demonstrated excellent detection capability, achieving an mAP0.5 of 77.47%, meeting the requirements for power tower detection in satellite remote sensing images. In comparison to Faster R-CNN, SSD, and RetinaNet, the LSKF-YOLO model exhibited significant performance gains with relatively smaller values for params and Flops. When comparing different YOLO models, they exhibited varying complexities and detection accuracies. YOLOv7-Tiny had the fewest parameters at 12.3 M, while LSKF-YOLO achieved the highest detection accuracy with model complexity within an acceptable range, with Flops and parameters at 22.6 G and 23.2 M, respectively.

Fig. 12 depicts the comparison of power tower detection results between YOLOv7-Tiny, Gold-YOLO, YOLOv5-S, and LSKF-YOLO. Based on the detection results from the SRSPTD dataset, YOLOv5-S showed instances of missed detection for distribution towers near roads, with lower detection accuracy.

For transmission tower detection, it occasionally misidentified trees as transmission towers. The detection accuracy of

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Fig. 13. Detection results of LSKF-YOLO on SRSPED dataset. (a) Detection result of power tower in forest plain. (b) Detection result of power tower in built environment.

YOLOv7-Tiny and Gold-YOLO is inferior to that of the model proposed in this article. In contrast, LSKF-YOLO exhibited reduced instances of false positives and false negatives compared to the baseline model, YOLOv7-Tiny and Gold-YOLO, with improved detection accuracy. The experimental results demonstrated that the adopted AFF-LSKM mechanism and SPD-MFAF structure effectively balanced the feature information of power poles at different scales, reducing the possibility of false target interference.

Fig. 13 showcases the detection results of LSKF-YOLO on the SRSPTD dataset. As can be seen from the figure, the network proposed in this article has the capability to effectively locate and identify transmission and distribution towers in satellite remote sensing images. However, there are limitations to consider. For instance, in the figure, the network struggles to differentiate trees with shapes similar to poles and towers, leading to false detections as indicated by the yellow circle. Furthermore, the network also demonstrates a failure to detect some distribution towers in residential areas, indicating a need for improved detection accuracy, as highlighted by the orange circle.

E. Discussion

The proposed network in this article is based on the concept of large convolutional kernel attention, utilizing the feature alignment fusion module to effectively fuse low-level feature information and achieve multiscale power pole detection.

The AFF-LSKM module decomposes the deep convolutional sequence to construct a large convolutional sequence and calculates attention, effectively utilizing contextual information in satellite remote sensing images. The SPD-MFAF module combines multiple low-level features to enhance high-level features, increasing feature dimension and information. For the detection of small targets such as distribution towers in satellite remote sensing images, the algorithm's performance

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surpasses the aforementioned comparative algorithms. However, due to the limited pixels occupied by distribution towers in the images, their low resolution, and their frequent location in densely populated areas such as residential zones, missed detections occur due to difficulties in distinguishing them from the surrounding environment. In addition, some distribution towers and streetlights have similar shapes, leading to false detections. Therefore, the proposed model may face challenges in ensuring that the constructed features are sufficiently interpretable and discriminative for detecting small distribution tower targets. The contribution of the distribution tower loss to the overall model loss during training is relatively small, resulting in the suboptimal effectiveness of the extracted features. As deep learning techniques continue to be applied in super-resolution technology, small target detection methods based on super-resolution have strong interpretability and have achieved impressive results. In future work, enhancing the feature detection of small targets using super-resolution technology will be a focal point of our research. In addition, the direction of power lines in satellite remote sensing images is closely related to the positions of point towers. We will comprehensively utilize the direction information of power lines to construct a multitask detection model, further improving the detection performance of transmission and distribution infrastructure in satellite remote sensing images. In addition, the exchange of repetitive information between images with different resolutions and the transfer of learning knowledge under different backgrounds will also be a problem that we need to pay attention to in the future. At the same time, the spectral characteristics of satellite remote sensing images reveal the potential information and hidden value of ground objects, and the fusion algorithm of space and spectral information preservation using hyperspectral remote sensing data will also be the focus of our future research.

V. Conclusion and Future Work

The article introduces the LSKF-YOLO algorithm for detecting power towers in high-resolution satellite remote sensing images. The algorithm incorporates a large kernel space selective attention fusion module, an MFAF module, and introduces MPDIoU to improve CIoU, effectively enhancing algorithm accuracy. First, the addition of the AFF-LSKM structure to the backbone feature extraction layer enlarges the network's RF and strengthens feature extraction capability. Second, the MFAF structure in the feature fusion layer enhances feature representation in the network and effectively utilizes low-level semantic information. Finally, replacing the step convolution of the last two layers of the neck network with SPD-Conv enhances the detection ability of small targets such as distribution towers in the image. Experimental results on SRSPTD data demonstrate the method's strong performance in detecting power tower targets in satellite remote sensing under complex backgrounds.

There are some limitations in this study, which can be improved in the future work. The main points of future research and suggestions for improvement are listed as follows.

1) Utilize super-resolution techniques to enhance the features of small targets such as distribution towers, thereby improving detection accuracy.
2) Integrate the directional information of power lines within images and develop a multitask detection model by combining pixel-level power line detection, enhancing the detection performance of transmission and distribution infrastructure in satellite remote sensing images.
3) Multispectral remote sensing data, particularly hyperspectral imagery, offers significant advantages in target detection. In future research, we plan to combine hyperspectral data to study a spatial–spectral dual optimization coupling model, fully utilizing satellite remote sensing images of power towers, and explore ways to solve spectral variability.
4) To address the issue of varying resolutions and differences in satellite remote sensing images under different environments, we aim to explore methods to integrate high and low-resolution data, reduce interimage discrepancies, and facilitate the exchange of repetitive information across resolutions and the transfer of knowledge under diverse backgrounds.

Broadly speaking, by combining diverse remote sensing data, enhancing feature learning methods, and integrating domain expertise, we aim to develop a detection model for high-resolution satellite remote sensing images driven by data and knowledge. This approach is intended to enhance the model's performance and robustness in detecting transmission and distribution infrastructure.

VI. Data Availability

The dataset used in this study is supported by the Bass Connections Project of Duke University, USA. The datasets can be accessed from https://figshare.com/articles/Electric_Transmission_and_Distribution_Infrastructure_Imagery_Dataset/6931088.

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Chaojun Shi (Member, IEEE) was born in Hebei, China, in 1992. He received the Ph.D. degree from Hebei University of Technology, Tianjin, China, in 2021.

He is currently a Lecturer with North China Electric Power University, Baoding, China. He has published about ten papers in international journals and conferences, such as the IEEE GEOSCIENCE AND REMOTE SENSING LETTERS, *International Journal of Remote Sensing*, and *IET Signal Processing*. His research interests include remote sensing computer vision and deep learning.



Xian Zheng (Student Member, IEEE) was born in Guangxi, China, in 2000. She is currently pursuing the master's degree with North China Electric Power University, Baoding, China.

Her research interests include remote sensing computer vision, power vision, and deep learning.



Zhenbing Zhao (Member, IEEE) was born in Jiangsu, China, in 1979. He received the B.S., M.S., and Ph.D. degrees from North China Electric Power University, Baoding, China, in 2002, 2005, and 2009, respectively.

He is currently a Professor with the School of Electrical and Electronic Engineering, North China Electric Power University. His research interests include artificial intelligence, intelligent detection of electrical equipment, and image processing.



Ke Zhang (Member, IEEE) received the Ph.D. degree from Beijing University of Posts and Telecommunications, Beijing, China, in 2012.

He is currently a Professor with North China Electric Power University, Baoding, China. He has published about 30 papers in international journals and conferences, such as IEEE TRANSACTIONS ON INSTRUMENTATION AND MEASUREMENT, *Frontiers in Neuroscience*, IEEE/CAA JOURNAL OF AUTOMATICA SINICA, IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS FOR VIDEO TECHNOLOGY, and *Neuro Computing*. His research interests include deep learning, computer vision, power vision, and biometrics.

Zibo Su is currently pursuing the master's degree with North China Electric Power University, Baoding, China.

His research interests include ground-based cloud segmentation, photovoltaic power prediction, and deep learning.

Qiaochu Lu is currently pursuing the master's degree with North China Electric Power University, Baoding, China.

Her research interests include power vision and deep learning.

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